EXP 4

Create arrays from vectors with dimension names, print specific elements.

Aim:

To create arrays from vectors with dimension names, print specific elements.

Procedure:

Step 1: Open RStudio or R console on your system.
Step 2: Create a sequence of numbers and store it in a variable.
Step 3: Use the array function to arrange these numbers into multiple dimensions by specifying the number of rows, columns, and tables.
Step 4: Store the created array in a variable.
Step 5: Display the entire array to view how the elements are arranged in each layer.
Step 6: Access a specific element from the array using its row, column, and table position.
Step 7: Print the selected element to verify correct indexing within the array.

Code:

```
data <- 1:24
arr <- array(data, dim = c(3, 4, 2),
             dimnames = list(
               Rows = c("R1", "R2", "R3"),
               Columns = c("C1", "C2", "C3", "C4"),
               Tables = c("T1", "T2")
             ))
cat("3D Array:\n")
print(arr)
cat("\nElement at R2, C3, T1:\n")
print(arr["R2", "C3", "T1"])
```

Output:

```
, , Tables = T1

    Columns
Rows C1 C2 C3 C4
  R1  1  4  7 10
  R2  2  5  8 11
  R3  3  6  9 12

, , Tables = T2

    Columns
Rows C1 C2 C3 C4
  R1 13 16 19 22
  R2 14 17 20 23
  R3 15 18 21 24


Element at R2, C3, T1:
[1] 8
```